

SUPPLY CHAIN VISIBILITY & OPTIMIZATION

MILESTONE 4 – DETAILED PROJECT DOCUMENT

Warehouse Efficiency Reports • Executive Overview Dashboard •
Optimized Dashboard Performance • Final Testing, Documentation
& Deployment

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Abstract

This document presents Milestone 4 of the Supply Chain Visibility & Optimization project. The milestone focuses on converting the earlier data analysis and Power BI work into a more complete, practical, validated, and deployment-ready supply-chain solution. The work covers warehouse efficiency reporting, an executive overview dashboard, dashboard performance optimization, and final testing, documentation, and deployment. The overall objective is to improve visibility, identify performance gaps, support operational decisions, and present supply-chain information in a clear and usable form.

1. Introduction

A supply chain covers planning, sourcing, manufacturing, transport, and delivery. Effective supply-chain visibility allows an organization to monitor inventory, suppliers, logistics, fulfillment, and other operational activities. Analytics helps transform these data points into meaningful information by identifying bottlenecks, controlling costs, and supporting better decisions.

In this project, Python is used for data preprocessing and data-quality preparation, while Power BI is used for interactive dashboards and performance analysis. The project journey progresses from clean data in Milestone 1, to analysis and dashboards in Milestone 2, advanced performance dashboards in Milestone 3, and finally the development, optimization, testing, documentation, and deployment activities of Milestone 4.

2. Problem Statement

Supply-chain operations can face several interconnected challenges. Shortages and excess inventory can affect availability and working efficiency. Delivery delays and logistics inefficiencies can increase operational pressure. Higher costs, inefficient planning, supplier and product quality issues, and reduced customer satisfaction can further impact overall performance.

- Stock: shortages and excess inventory.
- Delivery: delays and logistics inefficiencies.
- Cost: higher operational expenses.
- Planning: inefficient inventory planning.
- Quality: supplier and product quality issues.
- Customer: reduced customer satisfaction.

Aim: Analyze supply-chain performance, identify inefficiencies, and support data-driven optimization.

3. Project Objectives

- Understand and analyze the supply-chain dataset.
- Identify and resolve data-quality issues using Python.
- Perform exploratory data analysis to identify patterns and trends.
- Develop interactive Power BI dashboards for supply-chain analysis.
- Evaluate supplier, transportation, route, and carrier performance.
- Combine key performance indicators to generate actionable optimization insights.

4. Dataset and Earlier Milestones

The project dataset contains 1,540 records and 24 attributes. It provides information across product, logistics, supplier, manufacturing, inspection, and quality-related areas.

Important dataset areas include:

- Product: type, SKU, price, availability, products sold, and revenue.
- Logistics: shipping time, carrier, shipping cost, transportation mode, and route.
- Supplier and Manufacturing: supplier, manufacturing cost, lead time, inspection, and defect rate.

4.1 Milestone 1

Milestone 1 established a clean and reliable dataset. The process included importing the raw dataset, checking missing values and duplicates, reviewing data types and formatting, cleaning and standardizing the data, validating the results, and preparing the final dataset. Exploratory data analysis was then performed using univariate, bivariate, and multivariate analysis, statistical analysis, visualization, pattern identification, and business insight generation.

4.2 Milestone 2

Milestone 2 used EDA and Power BI to analyze inventory performance, category and product performance, fulfillment, and variance. The dashboards monitored overall inventory, revenue, shipping, and fulfillment performance and compared product, supplier, quality, and transportation performance.

4.3 Milestone 3

Milestone 3 introduced advanced dashboards for supplier scorecards, transportation cost analysis, route and carrier performance, and advanced KPI visualization. The analysis emphasized multi-metric supplier evaluation, transportation cost drivers, cost-time trade-offs, and integrated KPIs.

5. Milestone 4 – Developing, Optimizing, Testing and Deployment

Milestone 4 transforms the earlier analysis into a validated, documented, refined, and deployment-ready supply-chain solution. Four major activities are followed: Validate, Document, Refine, and Deploy.

5.1 Validate

The validation stage focuses on turning every KPI and visual into a verified result. The objective is to ensure that the final dashboard is accurate, consistent, and reliable.

5.2 Document

The documentation stage converts the entire project journey into clear evidence. It captures the methods, metrics, dashboards, analysis, and insights developed throughout the project.

5.3 Refine

The refinement stage transforms complex analysis into a simple decision-making experience. It focuses on clarity, usability, navigation, and visual impact so that users can understand the information more easily.

5.4 Deploy

The deployment stage moves the solution from project work toward practical delivery. The goal is to present a ready-to-use dashboard that supports smarter supply-chain decisions.

6. Milestone 4 – Warehouse Efficiency Reports

The Warehouse Efficiency Reports provide a broader view of operational performance by connecting inventory, fulfillment, orders, delivery, revenue, and defect information across locations. This dashboard is intended to help identify efficiency gaps and support data-driven supply-chain decisions.

6.1 Warehouse Efficiency Report – Dashboard I

The first warehouse-efficiency dashboard focuses on comparing inventory efficiency, revenue, defect rates, and fulfillment performance by location and key supply-chain dimensions.

The dashboard shows:

- Inventory performance across locations.
- Fulfillment performance.
- Order-related information.
- Delivery performance.
- Revenue performance.
- Defect performance.

Analysis focus: Compare inventory efficiency, revenue, defect rates, and fulfillment performance by location and other important supply-chain dimensions.

Purpose: Provide a centralized view to identify efficiency gaps and support data-driven supply-chain decisions.

6.2 Warehouse Efficiency Report – Dashboard II

The second warehouse-efficiency dashboard provides a more detailed operational view. It compares warehouse efficiency across locations and allows the analysis to drill down by product type, supplier, transportation mode, and shipping time.

The dashboard shows:

- Warehouse efficiency.
- Shipping costs.
- Product availability.
- Location-level performance.

Analysis focus: Identify location and supply-chain factors affecting warehouse efficiency for better operational decisions.

7. Milestone 4 – Supply Chain Executive Overview

The Supply Chain Executive Overview provides an executive-level view of overall supply-chain performance. It connects sales, revenue, production, manufacturing, shipping, quality, and location information in one view.

7.1 What the Executive Dashboard Shows

- Overall sales and revenue performance.
- Production volume.
- Manufacturing performance and cost.
- Shipping performance.
- Quality and defect information.
- Location-level performance.

7.2 Analysis Focus

The dashboard compares product types, carriers, locations, shipping time, revenue, production volume, manufacturing cost, and defect rate. These comparisons help management identify performance gaps and understand the major factors influencing supply-chain performance.

7.3 Purpose

The main purpose is to provide an executive-level view of key supply-chain KPIs so that decision-makers can quickly identify performance gaps and support data-driven decisions.

8. Milestone 4 – Optimize Dashboard Performance

Dashboard performance optimization focuses on making the analysis more useful for decision-making. The dashboard examines cost-time relationships and compares suppliers, locations, transportation modes, and product types through interactive filters.

8.1 What the Dashboard Shows

- Shipping cost.
- Shipping time.
- Supplier performance.
- Transportation modes.
- Revenue.
- Location-level analysis.

8.2 Analysis Focus

The analysis examines relationships between cost and time and compares important supply-chain dimensions. Interactive filters allow users to focus on specific suppliers, locations, transportation modes, and product types.

8.3 Purpose

The purpose is to optimize dashboard-driven supply-chain decisions by identifying cost patterns, efficiency gaps, and key performance factors.

9. Final Testing, Documentation & Deployment

Final testing, documentation, and deployment complete the Milestone 4 workflow. Testing ensures that dashboard outputs are reliable and that the visuals and KPIs behave consistently. Documentation records the project methods, metrics, dashboards, findings, and outcomes. Deployment prepares the solution for practical use.

9.1 Testing

- Verify KPI values and calculations.
- Check that visuals display the intended measures and dimensions.
- Verify filters and interactive selections.
- Check consistency between related dashboard views.
- Review the dashboard for clarity and usability.

9.2 Documentation

The documentation captures the complete project journey, including the dataset, preprocessing, EDA, dashboard development, metrics, analysis focus, insights, recommendations, and Milestone 4 improvements.

9.3 Deployment

Deployment represents the transition from project development to a practical dashboard solution. The final dashboard is intended to provide executive and operational users with accessible information for smarter supply-chain decisions.

10. Overall Key Insights

Supplier Performance: Multi-metric supplier evaluation helps identify supplier strengths and improvement areas.

Transportation Cost: Mode-, location-, and carrier-level analysis helps identify major transportation cost drivers.

Route & Carrier Performance: Combining shipping time and cost supports balanced route and carrier selection.

Advanced KPI Visualization: Integrated KPIs provide a consolidated view of operational performance and supply-chain bottlenecks.

11. Key Findings and Recommendations

11.1 Key Findings

- Supplier performance varies across sales contribution, stock levels, product contribution, and logistics indicators.
- Transportation costs differ across modes, locations, and carriers, highlighting major cost drivers.
- Shipping speed and transportation cost together reveal important route and carrier trade-offs.
- Integrated KPIs provide a consolidated view of operational performance and supply-chain bottlenecks.

11.2 Recommendations

- Continuously monitor supplier performance using multiple performance indicators.
- Review high-cost transportation routes and modes to identify optimization opportunities.
- Select routes and carriers by balancing shipping cost with delivery speed.
- Use interactive KPI dashboards for ongoing supply-chain performance monitoring.

Overall focus: Improve visibility → identify bottlenecks → optimize decisions → monitor performance.

12. Conclusion

The Supply Chain Visibility & Optimization project progresses from raw data preparation to advanced analytical dashboards. Milestone 1 established a clean and structured dataset. Milestone 2 used EDA and Power BI to identify patterns and monitor operational KPIs. Milestone 3 evaluated supplier, transportation, route, carrier, and KPI performance to support optimization.

Milestone 4 extends this work by developing warehouse efficiency reports, an executive overview, an optimized performance dashboard, and a final testing, documentation, and deployment process. As a result, the analysis is transformed into a more validated, documented, refined, and deployment-ready supply-chain solution that supports reliable insights and smarter decision-making.